# SYLLABUS

*for*

## FINAL YEAR DEGREE COURSE IN INSTRUMENTATION ENGINEERING (2003 COURSE)

# 20

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## SEMESTER-I

| Sub Code | Subjects | Teaching Scheme | | | Examination Scheme | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Lectures | Tutorial | Practical | Paper | T/w | Practical | Oral | Total |
| 406261 | Process Instrumentation-I | 04 | - | 02 | 100 | 25 | 50 | - | 175 |
| 406262 | Project Engineering & Management | 04 | - | 02 | 100 | 50 | - | 50 | 200 |
| 406263 | Digital Control | 03 | - | 02 | 100 | 50 | - | - | 150 |
| 406264 | Elective-I | 04 | - | 02 | 100 | 25 | - | 50 | 175 |
| 406265 | Project Presentation I | - | - | 02 | - | 50 | - | - | 50 |
| 406266 | Project | - | - | 02 | - | - | - | - | - |
| | Total | 15 | - | 12 | 400 | 200 | 50 | 100 | **750** |

| **Elective - I (Subject Code-406264)** | |
|---|---|
| Biomedical instrumentation | Laser applications in Instrumentation |
| Advanced Control system | Instrumentation for environmental engineering |

## SEMESTER-II

| Sub Code | Subjects | Teaching Scheme | | | Examination Scheme | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Lectures | Tutorial | Practical | Paper | T/w | Practical | Oral | Total |
| 406267 | Process Instrumentation II | 04 | - | 02 | 100 | - | - | 50 | 150 |
| 406268 | Computer Techniques & Applications | 03 | - | 02 | 100 | 50 | - | - | 150 |
| 406269 | Industrial Automation | 04 | - | - | 100 | - | - | - | 100 |
| 406270 | Elective-II | 04 | - | 02 | 100 | - | - | 50 | 150 |
| 406271 | Project | - | - | 06 | - | 100 | - | 50 | 150 |
| 406272 | Industrial Visits | - | - | 02 | - | 50 | - | - | 50 |
| | Total | 15 | - | 14 | 400 | 200 | - | 150 | 750 |
| Total Term-I & Terms-II | | 30 | - | 26 | 800 | 400 | 50 | 250 | **1500** |

| **Elective - II (Subject Code-406271)** | |
|---|---|
| Filer Optic Instrumentation | Signal Processing in Bio-Medical Instrumentation |
| Power Plant Instrumentation | Process Modeling and Optimization |

# 406261: PROCESS INSTRUMENTATION -I

| Teaching Scheme (Hrs/weeks) | | Examination Scheme (Marks) | | |
|---|---|---|---|---|
| **Lectures** | **Practical** | **Paper** | **Practical** | **Term work** |
| **4** | **2** | **100** | **50** | **25** |

***Prerequisite****: Basics of control valves and controllers.*

**UNIT 1: CONTROL VALVE DESIGN**

Designing control valve for gas, vapor and liquid services: Valve sizing by ANSI/ISA 75.01 std. Valve capacity testing by 75.02.

Effect and remedies of cavitation and flashing. Control valve noise generation and remedies. High temperature and High Pressure service valves. Control valve selection. Actuator Design considerations.

**UNIT 2: PROCESS CHARACTERISTICS**

**Types of processes**- Dead time, Single multi-capacity, self- Regulating, non regulating. Interacting / non interacting, Linear / non linear. Process gain, Process time constant. Step analysis method for finding first, second and multiple time constants and dead time. Dynamic elements in control loop. Process simulator.

**UNIT 3: MULTI LOOP AND MULTIVARIABLE CONTROL SYSTEM**

Feedback–Feed forward control, Cascade control, Ratio control, Selective control Adaptive control, Dual mode control, Split range control - Theory and industrial applications.

**UNIT 4: ANALYSIS AND PROPERTIES OF SOME COMMON LOOPS**

Flow, level, temperature, level loops, Non linear controllers, SLPC - features, faceplate, functions, MLPC - features, faceplate, functions, SLPC and MLPC comparison, **Scaling:** types of scaling, examples of scaling converting mathematical equations to engineering equations to scaled or normalized equators.

**UNIT 5: INTELLIGENT CONTROLLERS**

Principle of working, comparison with conventional controller, Model Based controller-Self tuning and Model reference Adaptive controller, Optimal controller, Predictive controller, Manufactures and their specifications.

**UNIT 6: ADVANCE PROCESS CONTROL TECHNIQUES**

Introduction to Advance process control techniques- Fuzzy logic systems and Fuzzy Controllers. Artificial Neural networks and ANN controller, Model Predictive Controller (MPC), Dynamic Matrix Controller (DMC) Introduction to Statistical Process Control (SPC).

**LIST OF EXPERIMENTS**

1. Control valve design using any software package.
2. Finding dynamic elements for any process ($T_D$, $T_S$).
3. Analysis of Flow/level loop (any one).
4 Analysis of Temperature / Pressure loop (any one).
5 Study of Feedback-Feed forward control .
6. Study of Cascade control/Ratio control/ Selective control (any one) .
7 Use of SLPC for process control.
8. Advance process controller study (ANN/Fuzzy/MPC) .

**(Students are expected to perform min. eight experiments)**

**TERM WORK**

Assignments based on Control valve paper design or PLC Advance Functions.

**TEXT BOOKS**

1. Process control Systems-F.G Shinskey, TMH
2. Computer Based Industrial Control -Krishna Kant, PHI
3. Fundamentals of Process Control - Murrill ISA
4. Chemical Process Control- Stephanopoulos George, PHI
5. Applications concepts of Process Control- By Murril ISA

**REFERENCE BOOKS**

1. Handbook of Control valves –ISA
2. Process Instrumentation and control Handbook –Considine
3. Statistical Process Control ISA
4. Handbook of Instrumentation Andrews Volume I
5. Handbook of Instrumentation -Process control --B G. Liptak, Chilton.

## 406262: PROJECT ENGINEERING AND MANAGEMENT

| Teaching Scheme (Hrs/weeks) | | Examination Scheme (marks) | | |
|---|---|---|---|---|
| **Lectures** | **Practical** | **Theory** | **Term work** | **Oral** |
| 4 | 2 | 100 | 50 | 50 |

**UNIT I**

Concept study and definition of Project Engineering and Management, Type of Standards and its studies as applicable to instrumentation and control engineering, Basics of Project Management, Degree of Automation, Organization Structure, Interdepartmental, Inter organizational and Multi agency interaction involved in Project and their co ordination Project statement.

**UNIT II**

**Project engineering documents and drawing:** P & I diagram based on Process Flow Sheet, Material balance sheet and Temperature pressure sheet, Methods of tagging and nomenclature scheme based on ANSI / ISA std, specification sheets, Instrument index sheets.

**UNIT III**

Cable Engineering (Class of conductors, Types, Specification and Application), Selection of cables with respect to specific application, Cable identification schemes, Cable trays. Plant layouts and General arrangement drawing (Plans and Elevation), Isometric of instrument piping, Loop wiring diagrams, installation sketches of filed instrument, BOM and MBOM.

**UNIT IV**

**Procurement activities:** Vendor registration, Tendering and bidding process, Bid evaluation, Purchase orders, Vendor documents, drawing and reports as necessary at above activities. Construction activities: Site conditions and planning. Front availability, Installation and commissioning activities and documents require at this stage, On site inspection and testing (SAT), Installation sketches, Contracting, Cold Commissioning and Hot commissioning, Customer Acceptance Test (CAT), Factory Acceptance Test (FAT), Performance trials and final hand over.

**UNIT V**

**Control centers and Panels:** Types, Design, Inspection and Specification, Control room engineering, Intelligent Operator Interface (IOI).**Networking:** Hubs, Routers, LAN Cards and CAT Cable.

## UNIT VI PROJECT MANAGEMENT

Project Management, Planning and Scheduling Life cycle phases, Statement of work (SOW), Project Specification, milestone scheduling, Work breakdown structure. **Cost and estimation:** Types of estimates, pricing process, salary overheads, labor hours, materials and support costs. Program evaluation and review techniques (PERT) and Critical path method (CPM), S-curve concept and crash time concepts, software's used in project management; software features, classification, evaluation and implementation.

**PRACTICAL (Any 8 Experiments)**

1) Study of standards and symbols (ANSI / ISA Std.)
2) Study of specification sheets.
3) Development of Process & Instrument diagram of typical process.
4) Development of Loop Wiring diagram.
5) Cable scheduling.
6) GA and mimic diagram of a control panel.
7) Development of Bar charts for certain project.
8) Preparation of Inquiry. Quotation, Comparative statement. Purchase orders, SAT, FAT and CAT. Inspection reports for control panel / transmitter/ control valve / recorder.

9) Hands on experience for engineering management software such as MS Project, Primavera.

**TEXT AND REFERENCE BOOKS**

1) Applied instrumentation in process industries by Andrew & Williams (Gulf Problems).
2) Management system by John Bacon (ISA)
3) Process control Instrument Engineers Hand book by Liptak.
4) Project Management - A System Approach to Planning, Scheduling and Controlling by Harold Kerzner (Van Nostrand Reinhold Publishing)
5) Instrument Installation Project Management (ISA).

# 406263: DIGITAL CONTROL

| Teaching Scheme (Hrs/week) | | Examination Scheme (marks) | |
|---|---|---|---|
| **Lectures** | **Practical** | **Theory** | **Term work** |
| **3** | **2** | **100** | **50** |

**UNIT-1 REVIEW OF DISCRETE TIME SYSTEMS**

Modeling of Discrete time Control system, Sampling theorem, Z transform, Inverse Z transform, Properties, Concept of Pulse transform function. Concept of Stability in Discrete Control systems: Jury Stability Test, bilinear transformation, Effect of sampling on stability.

**UNIT-2 STATE SPACE ANALYSIS OF DIGITAL SYSTEMS**

State Description of Digital Control System, Systems with dead time, Solution of sate equation, Pulse transfer Function Realization, discrete state transition matrix, similarity transformation. Lypunov Stability analysis, Multivariable Systems.

**UNIT-3 DIGITAL CONTROLLER**

Approximation of Digital Controller from Continuous system.Design consideration of digital controllers. Synthesis formula of Controller design. Digital PID Controller, Velocity and Positional form of PID controller and its algorithm, dead beat Response and ringing of poles.

**UNIT-4 POLE PLACEMENT AND OBSERVER DESIGN**

Concept of Controllability, Observability. Stability improvement by state feedback, Useful Transformation in state space analysis and design. Design via Pole placement. State observer. State regulator design, State feedback with Integral control.

**UNIT-5 ADVANCED CONTROL STRATEGIES**

Effect of dead time on System Performance. Smith Predictor algorithm. Model predictive Control - direct matrix Control and Internal Model Control, Self-tuning Control, Hybrid Control.

**UNIT-6 SYSTEM IDENTIFICATION AND OPTIMAL CONTROL**

Modeling and Identifications, Identification for Control, ARX Model,

ARMA Model, Prediction Error Method. Box- Jenkin Method, Output Error Method, Least square technique Parameter Optimization, Optimal control, Quadratic performance index, State Regulator design through the lyapunov equation, Optimal State Regulator through the matrix riccati equation, optimal digital control systems, Linear quadratic control.

***REFERENCE BOOKS***

1. Gopal. M. "Digital control Engineering ", Wiley Eastern Ltd., 1989.
2. G. F. Franklin, J. David Powell, Michael Workman, "Digital control of Dynamic Systems", 3rdEdition, Addison Wesley, 2000.
3. K. Ogata, "Discrete Time Control systems", Prentice Hall, Second Edition, 2003.
4. Forsytheand W. and Goodall R.N., "Digital Control", McMillan, 1991.
5. M. Gopal "Digital Control and State Variable Method" Tata – McGrow Hill, Delhi, 1997.
6. Cotantine H. and Gary B. Larnord "Digital Control Systems", second edition McGraw hill international, 2002.
7. Lennart Ljung, "System Identification – Theory for Users", Prentice Hall, 1999.

**LIST OF EXPERIMENTS**

****Perform any eight experiments using MATLAB out of the following:***

1. Obtain the response of the first order / second order system to any two standard inputs.
2. Determine the range of sampling period for stability of the system.
3. Effect of Dead time on system performance
4. Design of discrete controller – by – controller synthesis formula.
5. Design of dead beat controller.
6. Stability analysis of discrete time control system.
7. Determine the controllability and observability of a system.
8. To study the model predictive control technology.
9. Determine the parameters of an ARX / ARMA model by means of least square technique.
10. Design of control system using pole placement technique.
11. To study the state regulator design.

# 406264(1): BIOMEDICAL INSTRUMENTATION

| Teaching Scheme (Hrs/weeks) | | Examination Scheme (marks). | | |
|---|---|---|---|---|
| **Lectures** | **Practical** | **Theory** | **Term work** | **Oral** |
| **4** | **2** | **100** | **25** | **50** |

**UNIT I BIOPOTENTIAL MEASUREMENT**

Electrode-Electrolyte interface, half-cell potential. Polarization-polarisable and non polarizable electrodes, Ag/AgCI electrodes, Electrode circuit model; motion artifact. Body Surface record my electrodes for ECG, EMG, and EEG. Internal Electrodes- needle and

wire electrodes. Micro electrodes- metal microelectrodes, Electrical properties of microelectrodes. Electrodes for electric stimulation of tissue. ***Biotransducers:*** Physiological parameters & suitable transducers for its measurements, operating principles & specifications for the transducers to measure parameters. Biosensors Thermister used for cardiac output measurement, Intra vascular fiber optic pressure Transducer: Gauge type Blood pressure transducers;

**UNIT 2 CARDIOVASCULAR SYSTEM**

Heart Structure, Cardiac Cycle. ECG Theory. ECG Electrodes, Electrocardiograph, vectorcardiograph. Analog Signal Processing of Biosignals, Amplifiers, Transient Protection, interference reduction, Movement Artifact Circuits. Active Filters, Rate Measurement, Averaging an Integrator Circuits. Transient Protection Circuits

**UNIT 3 CARDIOVASCULAR MEASUREMENT**

Heart Sounds. Phonocardiograph. Blood Pressure Measurement (invasive an Noninvasive), Blood Flow meters: Magnetic, Ultrasonic, Thermal **Convection Method:** Cardiac Output Measurement (dye dilution method), Plethysmography

**UNIT 4**

**Central Nervous System**: Brain & its parts, different waves from different parts of the brain, brain stem, cranium nerves, structure of neuron, Neuron muscular transmission, Electroencephalography, Invoked Response. EEC amplifier Biofeedback. **Classification of muscles**- Muscle contraction mechanism, Myoelectric voltage, Electromyography (EMG).

**UNIT 5 SPECIAL SENSES**

1. **Ear**: Mechanism of Hearing. Sound Conduction System. Basic Audiometer; Pure ton audiometer: Audiometer system Bekesy; Evoked response Audiometer system Hearing Aids.
2. **Vision**: Anatomy of Eye, Visual acuity. (Errors in Vision.) Slit Lamp, Tonometer ophthalmoscope. Perimeter

**UNIT 6 RESPIRATORY INSTRUMENTATION**

Natural Process of Breathing. $O_2$ and $CO_2$ Transport, Regulation of Breathing, Spirometers, airflow measurement, Oxygenators-Bubble Type, Membrane Type Gas Analyzers: Infrared gas analyzer. Oxygen analyzer. Nitrogen analyzer, and Ventilators. ***Electrical Safety*** : Significance of Electrical Danger, Physiological Effect of Current, Ground Shock Hazards, Methods of Accident Prevention. **Practical:** *Students are expected to perform minimum 8 practical from the list mentioned below.*

**LIST OF PRACTICAL**

1 To Study and Check Specifications of an ECG Recorder.
2 To Design and Implement basic ECG Calibrator.
3 To Measure Blood Pressure Using Sphygmomanometer Calibration of BP apparatus.
4 Study of Audiometer.
5 To Design a Clinical Thermometer.
6 To record/monitor heart sounds using Electronic Stethoscope.
7 To Develop a Photoplethysmography Sensor for Pulse Rate Measurement
8 To Develop a Flow Type Sensor Using Thermistor for expiratory volume Measurement
9 To Design and Implement an ECG Simulator
10 To Design a Notch Filter for Power Line frequency
11 To Design and Implement an ECG Amplifier
12 To Implement a Heart Rate Meter
13 To Study EEG/EMG
14 To Study Ophthalmic instruments

***Books***

1 Human Physiology- The Mechanism of Body Function By Vander. Sherman. TMH Ed. 1981.
2 Introduction To Biomedical Equipment Technology By Carr & Brown
3 Biomedical Instrumentation and Measurements By Cromwell. 2nd edition. Pearson Education.
4 Handbook of Biomedical Instrumentation By R. S. Khandpur. TMH
5 Biomedical Digital Signal Processing. Tompkins, PHI
6 Biomedical Instrumentation. Arumugam
7 Text book of clinical Ophthalmology- Ronald Pitts Crick, Pang K haw 2nd Edition, World Scientific publication.
8 ISBn - 981-238-128-7 US $70.

# 406264(2): INSRUMENTATION FOR ENVIORNMENTAL ENGINEERING

| Teaching Scheme (Hrs/weeks) | | Examination Scheme (marks) | | |
|---|---|---|---|---|
| **Lectures** | **Practical** | **Theory** | **Term work** | **Oral** |
| **4** | **2** | **100** | **25** | **50** |

**UNIT I**

Environmental definitions, constituents, biochemical cycles, causes of pollution, types of pollution and their, measurement, effects of pollution, different sensors for measurement of pollution, difference between offline and online measurment and continuous monitoring. Review of standard methods of pollution analysis, sampling operation devices and techniques as related to environmental engineering.

**UNIT 2**

**Environmental toxicology and hazards:** common Toxic agents their analysis and safety measures environmental regulations and standards. Environmental testing Dry heat Dry cold Damp Heat Salt Spray Dust Altitude testing Bumping testing Vibration testing Drop/Topple testing free fall testing and study of ISO 14001.

**UNIT 3**

**Air pollution analysis:** analysis of aerosols and monitoring of gaseous pollutant like $SO_2$, $H_2S$, NO-$NO_X$, CO-$CO_2$, ozone. $NH_3$, and organic gases, vapour analysis, monitoring of suspended particulate matter and trace metal pollutants. Instrumentation setup for air pollution analysis. HVAC control.

**UNIT 4**

**Water pollution analysis:** physical examination colour, conductivity, temperature, odour, turbidity, hardness. Chemical characterization *Ca*2 +, *Mg* 2 + *Na* + *CI* – $SO_4$+2-,*HCO*3,*AI*3 + *Ba*2+boron *F*-,*NO*2-,*PO*4 3- ,*Fe*3+, *Mn*2-,*Sio*2 2-, Biological investigation DO,BOD, bacteriological examination, types of water quality monitoring instruments, (pH meter, conductivity meter, etc). Instrumentation setup for waste water treatment.

**Effluent Analysis:** Physical method of characterization, BOD, COD, TOC, specific analysis of organic pollutants, analysis of trace metal pollutants, analysis of anion and dissolved gases, dissolved oxygen, pH, dissolved chlorides, suspended solids, nitrogen, and sludge index.

**UNIT 5**

**Noise Pollution unit Measurement:** Units, devices, and maps, noise control system. Effects of noise pollution and noise reduction techniques. Radiation Pollution: Radiation pollution, its measurement and control. Effects of radiation pollution on living and non living things.

**UNIT 6**

**Soil Pollution and Pesticide Analysis:** Analysis of micronutrients trace elements pesticides, chromatographic characterization, Polarographic and spectroscope analysis of pesticides. Instrumentation setup for soil pollution.

***LIST OF PRACTICALS***

1. Offline measurement of CO
2. Measurement of NO - $NO_X$.
3. Measure Hardness of Water.
4. Measurement of Dissolved Chloride.
5. Measurement of dissolved Oxygen.
6. Study of Sound level meter.
7. Measurement of COD, solids, conductivity and alkalinity.
8. Case Study

**TEXT BOOKS**

1. Environmental Pollution Analysis by S. M. Khopkar
2. Basic Environmental Technology by. J. A. Nathanson
3. Environmental technology by S. C. Bhatia
4. Environmental Science by R. Vcnugopal

**REFERENCE BOOKS**

1. R. K. Trivedy, S.N. Trivedy "Advances in Wastewater Treatment".
2. R. K. Trivedy, Arvind Knmar, “Encyclopedia for Pollution control and environmental analysis”.

# 406264(3): LASER APPLICATIONS IN INSTRUMENTATION

| Teaching Scheme (6 Hrs/week) | | Examination Scheme(marks) | |
|---|---|---|---|
| **Lectures** | **Practical** | **Paper** | **Oral** |
| **4** | **2** | **100** | **50** |

**UNIT 1: LASER FUNDAMENTALS**

Properties of lasers, Eienstein equations, population inversion, optical feedback and laser oscillation, threshold conditions for laser oscillation, lineshape functions. Laser modes-Axial and Transverse, single mode operation. Frequency stabilization. Mode locking. Mode hopping. Q-switching.

**UNIT 2: LASER TYPES AND LASER SAFETY**

**Classes of lasers:** Doped insulator lasers. Semiconductor lasers, Gas lasers. Liquid Dye lasers etc. **Laser safety:** Biological effects, safety standards, risk of exposure, laser hazard classification and assessment, laser safety system, safe industrial laser laboratory, laser eye protection, laser accidents.

**UNIT 3: LASER INTERFEROMETRY AND SPECKLE PATTERN INSTRUMENTS**

**Laser Interferometry:** Basic Optical Interferometers, Performance parameters. Ultimate limits of performance, Laser vibrometry- short distance, medium distance and long distance vibrometry. Injection Interferometry, white light Interferometry. **Speckle pattern instruments:** Speckle properties, speckle in single point interferometers and electronic speckle pattern Interferometry.

**UNIT 4: LASER DOPPLER VELOCIMETRY**

**Laser Doppler Velocimetry:** Principle of operation, performance parameters: Scale factor relative error, Accuracy of the Doppler Frequency, Size of sensing region, alignment and positioning errors etc. Electronic processing of the Doppler signal: Time domain and Frequency domain processing. Optical configurations.

**UNIT 5: LASER GYROSCOPES**

**Gyroscopes:** The Sagnac effect, Basic gyro configurations. Ring Laser Gyros (RLG): Dithered RLG, Ring Zeeman laser gyro, performance of RLGs. **Fiber Optics Gyros (FOG):** Open loop FOG, Requirements on FOG components, technology to implement FOG, Closed loop FOG, the resonant FOG MEMS gyro, Piezoelectric gyro

**UNIT 6: HOLOGRAPHY**

The basic principles of Holography, viewing a hologram, volume hologram, multiplex hologram, white light reflection hologram. Measurement of strain, stress, bending moments and vibration by Holography, nondestructive testing, medical and dental research, solid mechanics.

## *LIST OF EXPERIMENTS*

**Study of LASERS,**

**Aim:** Make a humble start in theoretical and practical understanding of lasers. The study is limited to continuous HeNe laser and red diode laser.(easy availability)

**A.** Following properties of he lasers are proposed to be studied

1. Current/intensity relationship of a diode laser.
2. Spectrum plot w.r.t. current and near lasing threshold for a diode laser.
3. Spectrum plot w.r.t. current and near lasing threshold for a HeNe laser.
4. Beam width and beam profile of a laser.
5. Measurement of polarization of a laser output, experiment on polarised light (plane, circular, elliptical etc.)

**B.** An attempt shall be made to measure coherence length and beam divergent angle.

**C.** Opto-electrical characteristics of laser diode- comlex impedance, rise time and fall time constants of a laser output, direct electrical modulation of laser intensity, frequency modulation, pulse-width modulation and pulse position modulation.

**D.** Michelson interferometer with laser.

**E.** Scope of the study can be further extended to pulse lasers, frequency doubling and other/related nonlinear phenomenon. A.YAG laser B. Optical fiber laser.

## REFERENCES

1. Laser electronics, Joseph T Verdeyen, Prentice-Hall of India Pvt. Ltd., second edition, -1993.
1. Electro-Optical Instrumentation, Silvano Donati, Pearson Education, Inc., 2004
2. Optical Fiber Communications. John M. Senior, Prentice-Hall of India Pvt. Ltd., second edition, 1994.
3. Holographic Interferometry, Charles M. Vest, John Wiley & sons, 1979.

# 406264(4): ADVANCED CONTROL SYSTEMS

| Teaching Scheme (Hrs/weeks) | | Examination Scheme (marks). | | |
|---|---|---|---|---|
| Lectures | Practical | Theory | Oral | Term work |
| 4 | 2 | 100 | 50 | 25 |

**UNIT-1 NON-LINEAR SYSTEMS**

Types of non-linearity, typical examples, singular points, Phase plane analysis, Limit cycles, linearization, Describing functions. Need for model reduction, Dominant pole concept. Model reduction via partial realization. Time moment matching and pade approximation, Hankel norm model reduction.

**UNIT-2 STABILITY**

**Stability concepts** - Equilibrium points - BIBO and asymptotic stability, Lyapunov Theory, Definitions (Stability and Functions). Direct method of Lyapunov, Application to non-linear problems. Stability analysis by describing function method -jump resonance. Frequency domain stability criteria, Popov's method and is extensions.

**UNIT-3 MODEL REFERENCE ADAPTIVE SYSTEMS**

Different configurations and classifications of MRAC - Mathematical description - Direct and indirect model reference adaptive control - MIT rule for continues time MRAC systems -Lypunov approach and hyper stability approach for continuous time and discrete time MRAC systems - Multivariable systems – Stability and convergence studies.

**UNIT-4 SELF TUNING REGULATORS**

Different approaches to self-tuning - Recursive parameter estimation - Implicit and explicit STR -LQG self-tuning - Convergence analysis - Minimum variance and pole assignment approaches to multivariable self-tuning regulators.

**UNIT-5 RECENT TRENDS AND APPLICATIONS OF ADAPTIVE CONTROL**

Recent trends in self-tuning Robustness studies multivariable system. Model updating. General-purpose adaptive regulator. Application to Process control components and systems. Industrial Applications.

**UNIT 6 OPTIMAL CONTROL**

Problem formulation, necessary conditions of optimality, state regulator problem. Matrix Riccati equation, infinite time regulator problem, output regulator and tracking problems. Pontryagin’s minimum principles, time, and optimal control problem. Dynamic programming. Linear Quadratic Regulator, model matching based on Linear Quadratic optimal regulator. Observer design, Linear optimal filter.

**REFERENCE BOOKS**

1. Chalam, V.V., "Adaptive Control Systems", Techniques & Applications, Marcel Dekker, Inc. NY and Basel. 1987.
2. Eveleigh, V.W., "Adaptive Control and Optimisation Techniques". McGraw-Hill, 1967.
3. Narendra and Annasamy, "Stable Adaptive Control Systems", Prentice Hall, 1989.
4. Astry, S. and Bodson, M., "Adaptive Control", Prentice Hall,1989.
5. M. Vidyasagar, "Nonlinear Systems Analysis", 2nd Ed., Prentice Hall, 1993.
6. Hassan K. Khalil, "Nonlinear Systems", Third Edition, Prentice Hall, 2002.
7. William S. Levine (Editor), "The Control Handbook(Electrical Engineering Handbook Series)", CRC Press, March 1996.
8. Nagrath I.J., and Gopal, M., "Control system Engineering" Wiley Eastern Reprint 1995.
9. Kirk D.E., "Optimal control theory-an introduction", Prentice Hall, N.J. 1970.
10. Gopal. M., "Modern control system Theory", Wiley Eastern Ltd., 2nd Edition Reprint 1995.
11. Graham C., Goodwill, S. F. Graebe and M. E. Salgado,"Control
12. System Design" Prentice Hall India, New Delhi, 2002.

**LIST OF EXPERIMENTS**

** Perform any eight experiments using MATLAB out of following*

1. Analysis of first order/second order non-linear system.
2. Effect of Dominant pole and Critical pole on system performance.
3. Stability analysis of first order/ second order system by describing function method.
4. Obtain the stability of a system by Frequency domain criteria.
5. Study of Direct/indirect model reference adaptive control system.
6 Study of multivariable self-tuning regulators.
6. Analysis of Multivariable systems using step input
7. Any one Industrial Application of model reference control-a Survey.
8. Design of state observer
9. Design of linear filter.

# 406265: PROJECT PRESENTATION (TERM-WORK: 50 MARKS)

*Under this head, it is suggested that student should identity their project as per the requirement of part II of the syllabus and study the literature required for this project. The student should present progress of the proposed project. This progress presentation consists of problem statement, literature survey, project overview and scheme of implementation (block diagram, PERT chart etc.). All students are expected to make an audio-visual presentation of their study and submit relevant documents as term-work.*

# 406267:PROCESS INSTRUMENTATION - II

| Teaching Scheme (Hrs/week) | | Examination Scheme(marks) | |
|---|---|---|---|
| Lectures | Practical | Paper | Oral |
| 4 | 2 | 100 | 50 |

**Prerequisite:** ***Process Plant operations***

**UNIT 1 HEAT EXCHANGERS AND CHILLERS**

Basic & multi variable instrumentation and control schemes including safety and performance considerations for Heat Exchangers and Chillers.

**UNIT 2 CHEMICAL REACTORS**

Basic & multi variable instrumentation and control schemes including safety and performance considerations for Batch Reactors and CSTRs.

**UNIT 3 PUMPS AND COMPRESSORS**

Basic & multivariable instrumentation and control schemes including safety and performance considerations for Pumps and Compressors

**UNIT 4 BOILER**

Basic & multivariable instrumentation and control schemes including safety and performance considerations for Boiler.

**UNIT 5 DISTILLATION COLUMN**

Basic & multivariable instrumentation and control schemes including safety and performance considerations for Distillation Column.

**UNIT 6 CRYSTALLIZERS, EVAPORATORS, DRYERS**

Basic & multivariable instrumentation and control schemes including safety and performance considerations for Crystallizers, Evaporators, Dryers

***LIST OF EXPERIMENTS***

*Study of various process plants w.r.t. Applicable instrumentation & Control schematics for Supervisory, Modulating, Safety & sequencing operations.*

1. Boiler Equipments
2. Heat Exchangers
3. Compressors
4. Pumps
5. Evaporator
6. Crystlizer
7. Dryers
8. Development of P & Id for complete Batch process reactors as per Sequence of operations described in assignment sheet.
9. Study of Process Plant in association with DCS – SCADA applications, Instrumentations & Control Schematics for Supervisory, Modulating, Safety & sequencing operations.
10. Study of Manufacturing Plant in association with PLC - SCADA

applications. Instrumentations & Control Schematics for Supervisory, Modulating, Safety & sequencing operations.

*Students shall prepare technical descriptive part of each items Respective description supported with the concerned engineering drawing/sketch, SAMA diagrams SFC flow chart.*

***(Student's are expected to perform min. eight experiments)***

***TEXT BOOKS***

1 Process control Systems- F.G.Shinskey,TMH
2 Computer Based Industrial Control —Krishna Kant ,PHI
3. Chemical Process Control-Stephonopolous, PHI

***REFERENCE BOOKS***

1. I.Batch Control System-T.G.Fisher, lSA
2. Distillation Column Control -F.G. Shinskey, TMH
3. Boiler Control System -Lindsley D.,TMH
4. Handbook of instrumentation -Process control —B.G. Liptak, Chilton.

## 406268: COMPUTER TECHNIQUES & APPLICATIONS

| Teaching Scheme (Hrs/week) | | Examination Scheme(marks) | |
|---|---|---|---|
| **Lectures** | **Practical** | **Theory** | **Term work** |
| **3** | **2** | **100** | **50** |

***Prerequisites:*** Students must be good at **"C"** programming and logic development.

**UNIT 1**

Concepts of Operating System and its services, Types of operating systems. Process Management: Concept, scheduling, operations on process . CPU scheduling: Basic concepts, CPU scheduling algorithms . Deadlocks Characterization. Handling. Recovery . Disk scheduling algorithms.

**UNIT II**

**Memory Management:** Address Binding, Overlays, Swapping. Contiguous memory allocation. Paging, Segmentation

Virtual memory Concept. Demand paging. Prepaging. Page size considerations. Page Replacement algorithms. Thrashing

File system management : Concept file access methods, directory structures, file allocation methods.

**UNIT III**

**Parallel computers:** Basic concepts. Type of parallelism. Intertask dependencies, classification of parallel computers, vector computers. Array processors. Systolic arrays RTOS concepts. Types scheduling algorithms.

Data compression: Overview of Information Theory, Huffman Coding.

Loss less and lossy compression.

**UNIT IV**

**Computer Communication:** ISO-OSI seven layer model. The TCP/IP reference model Introduction to LAN. LAN topologies. IEEE standards for networking- IEEE 802.3, 1EEE 802.4, IEEE 802.5, Circuit switching and Packet switching networks. Features and capabilities of TCP/IP, Industrial Ethernet

**UNIT V**

**Introduction to IEEE 1394. IEEE488(GP1B) :** its configuration and advantages. Introduction to ARM processors and features of ARM9 and ARM 10E core families. Software quality concepts. Software reliability, ISO 9000 and SEI standards for software.

**UNIT VI**

**Software Development Life Cycle:** Software Testing, fundamentals, white box. Black box testing, control structure testing, strategic approach to testing, unit testing, integrated testing, validation testing, system testing, CASE tools.

***LIST OF PRACTICALS***

1. CPU scheduling algorithms.
2. Huffman Coding.
3. PC to PC Communication.
4. Programming in C covering Functions/ Arrays, eg: Calculator.
5. Programming in C covering Input/ Output (Console or Disk).
6. Process simulation using C or C++ Graphics.
7. Writing program in VB for familiarization with User Interface.
8. Writing program in VB for data base connectivity.
9. Writing program in VB using timers.

***(Note: Perform any 8 practical for the above list)***

***LIST OF BOOKS***

1. Operating System Concepts by Silberschatz, Galvin Gagne.
2. Advanced MS DOS programming by Ray Duncan. 2nd edition, bpb publications
3. Parallel Computer architecture and programming by V. Rajaraman. C. Siva Ram Murthy, PHI
4. Computer Architecture and Parallel processing by Kai Hwang Faye briggs McGraw Hill international editions
5. Computer Networks Protocols, Standards and Interfaces by Uyless Black, PHI.
6. Computer Networks by Andrew Tanenbaum, Prentice Hall.
7. Introduction To Data Compression by Khalid Sayood. Morgan Kaufmann Publishers, Inc.
8. High Speed Networks TCP/IP and ATM design principles by William

stalling.
9. Software Engineering, A practitioner's Approach, 6th edition McGraw Hill international editions.
10. Software Engineering by lan Somerville, 4th edition, Addison Wesley publication
11. www.interfacebus.com
12. www.arm.com

# 406269: INDUSTRIAL AUTOMATION

| Teaching Scheme | Examination Scheme |
|---|---|
| Lectures | Paper |
| 4 | 100 |

**UNIT I PLANT WIDE CONTROL SYSTEMS AND AUTOMATION STRATEGY**

Evolution of instrumentation and control, Role of automation in industries. Benefits of automation. Introduction to automation tools PLC, DCS, SCADA, Hybrid DCS/PLC, Automation strategy evolution. Control system audit, Performance Criteria, Development of User Requirement Specifications (URS) for automation, Functional Design Specifications (FDS) for automation tools.

**UNIT 2 ADVANCE APPLICATIONS OF PLC**

PLC programming methods as per IEC 61131, PLC applications for hatch process using SFC, Analog Control using PLC. PLC interlace to SCADA/DCS using communication links (RS232, RS485) and protocols (Modbus ASCII/RTU).

**UNIT 3 INSTRUMENTATION STANDARD PROTOCOLS**

HART Protocol introduction, frame structure, programming, implementation examples, Benefits, Advantages and Limitations. Foundation Fieldbus HI introduction, structure, programming, FDS configuration, implementation examples, Benefits, Advantages and Limitations. Comparison with other fieldbus standards including Devicenet, Profibus, Controlnet, ASL industrial Ethernet etc.

**UNIT 4 DISTRIBUTED CONTROL SYSTEMS BASICS**

DCS introduction, functions, advantages and limitations, DCS as an automation tool to support Enterprise Resources Planning, DCS components/ block diagram. Architecture of different makes, DCS specifications. Latest trends and developments , Performance Criteria for DCS and other automation tools.

**UNIT 5 DISTRIBUTED CONTROL SYSTEMS ENGINEERING AND DESIGN**

DCS detail engineering, configuration and programming, functions

including database management, reporting, alarm management, communication, third party interface, control, display etc. Enhanced functions viz. Advance Process Control. Batch application. Historical Data Management, OPC support, Security and User Access Management.

**UNIT 6 APPLICATION DEVELOPMENT AND AUTOMATION FOR INDUSTRY VERTICALS**

**Other Automation Tools -** Introduction and Applications of safety systems, SCADA Application development and Automation for following industries. **Batch Processes:** Chemical/ Pharmaceutical, Food. **Continuous Processes:** power . Paper & Pulp. Cement.

***BOOKS / REFERENCES***

1. The Management of Control System: Justification and Technical Auditing N.E. Battikha, ISA.
2. Computer Aided Process Control S.K. Singh, Prentice Hall of India
3. Programmable Logic Controllers: Principles and Applications Webb & Reis, Prentice Hall of India
4. Introduction of programmable Logic Controllers ,Garry Dunning, Thomson Learning
5. Hatch Control Systems. Design. Application and Implementation Thomos Fisher. ISA
6. Distributed Control for Industrial Automation Poppovik Bhatkar, Dekkar Publication computer Based Process Control Krishna Kant. Prentice Hall of India.
7. www.hartcomm.org.
8. www.fieldbus.org

# 406270(1): SIGNAL PROCESSING IN BIOMEDICAL INSTRUMENTATION

| Teaching Scheme (Hrs/week) | | Examination Scheme(marks) | |
|---|---|---|---|
| **Lectures** | **Practical** | **Theory** | **Oral** |
| **4** | **2** | **100** | **50** |

**UNIT I**

**Life Saving Devices** : Pacemaker, Types of pacemakers: External & Internal, **Defibrillators:** AC & DC Defibrillator, Heart Lung Machine, **Elements of Intensive Care Monitoring:** Bedside Monitors. Drug Delivery System, ICU layout. **Operating Room instrumentation**: Electro surgical Unit, Anesthesia Machine.

**UNIT 2**

**Clinical Lab Instrumentation:** Blood and its composition and function.

Electron Microscope, Blood Cell Counters, Electrophoresis, Pulse Oximetry, Conventional and Automated, Hb and Glucose Measurement. Introduction to telemetry & Telemedicine.

**UNIT 3**

**Imaging Systems:** X ray properties. Generation of X-rays, block diagram of x Ray machine, image intensifier. Draw back of x-ray imaging, CT Scanning, basic CT Scanning system. Types of gantries, gray scale [Hounsfield No.], image reconstruction techniques in tomography, image artifacts.

**UNIT 4**

**Advanced imaging Systems:** Radionuclide Imaging: Rectilinear Scanner. Scintillation Camera. Positron Emission Tomography, Single Photon Emission Computed Tomography. Ultrasound Imaging: Fundamentals of Acoustic propagation. Ultrasonic transducers and frequencies, A. B. M Scan and Echocardiography, Introduction to MRI &V thermography.

**UNIT 5**

**Laser applications in Medicine:** Types of Lasers, Properties of Laser. Interaction of lasers with tissues -Thermal and Non thermal, Basic Endoscopcs system & its characteristics. Laser Applications in ophthalmology- Diabetic Retinopatln , Glaucoma and Retinal hole and detachment treatment, Dermatology- Tattoo, port wine treatment. **Pain relief Instrumentation Diathermy**: short wave. Microwave, Ultrasound diathermy

**UNIT 6**

**Concept of Rehabilitation Engineering:** Orthrotics & Prosthetic devices, overview of various orthrotics & prosthetic devices along with its materials. Wheelchair -Types, Materials used in wheelchair, Joysticks used in wheelchair. **Kidney Instrumentation:** Kidney Structure, Regulation of Water and Electrolyte Balance, Artificial Kidney-types (Coil type, parallel plate Type), Dialysis System, Lithotripsy.

***PRACTICALS***

*Students are expected to study minimum 8 equipments by visiting clinics/hospitals*

***List of equipments***

1 Endoscope
2 Bedside Monitor, Drip Rate Monitor (ICU Instrumentation)
3 Electrosurgical Unit (Operating Room)

4 Various Imaging Techniques
5 Pacemaker & Defibrillator
6 Clinical Lab Instruments
7 Short Wave Diathermy
8 Dialysis equipment
9 Clinical Lab instrumentation
10 ECG Telemetry System
11 Rehabilitation equipments
12 Diabetic Retinopathy Treatment using laser

***Books***

1. Medicine and Clinical Engineering By Jacobsons & Webster, PHl
2. Introduction To Biomedical Equipment Technology By Carr & Brown
3. Biomedical Instrumentation and Measurements By Cromwell, PHI
4. Handbook of Biomedical Instrumentation By R. S. Khandpur, TMH
5. The Biomedical Engineering Handbook, Bronzino, IEEE Press
6. Applied Chemical Engineering -Feenberg,
7. Principles of Medical Imaging By: K. Kirk Shung, Michael B. Smith, Benjamin Tsui. Pub: Academic Press.
8. Medical Laser Applications By Carruth
9. Medical Lasers & their sale Use By Sliney & Trokal.

**(Elective II)**

## 406271 (2): POWER PLANT INSTRUMENTATION

| Teaching Scheme (Hrs/weeks) | | Examination Scheme (marks) | |
|---|---|---|---|
| **Lectures** | **Practical** | **Paper** | **Oral** |
| **4** | **2** | **100** | **50** |

**UNIT 1. Energy:**

Forms of energy and it's major sources. Importance of electricity as source of energy. Fundamentals of generation of Electricity, its transmission and Distribution. Concept of regional and national power grid. Concept of distance protections and islanding. Comparison of thermal power plant, hydroelectric power station and nuclear power plant as main sources of generation of electricity.

**UNIT 2. Thermal and Gas Turbine Power Plant:**

**A) Thermal Power Plant:** Unit overview, types of Boilers. Construction of pulverized fuel fired water tube boilers. Air and Flue gas paths. Accessories like Fans, Pumps, Coal Mills and Furnace oil handling systems. Feed water and steam cycles. Burner management system. Sequence starting system of auxiliaries and safety considerations. Coal and Fly ash handling systems. Instrumentation and controls for water

treatment and demineralised water plant. Automatic reactivation system for resin beds. Selection of control valves for steam, oil and feed water.

**B) Gas Turbine:** Block diagram of Gas Turbine generator. Waste heat recovery boiler.

**UNIT 3. Instrumentation and Control for Turbine and Generator:**

Hydraulically controlled speed governing and turbine steam inlet control valve actuation system. Thermal stress measurement and control. Measurement of Vibrations, Differential Expansions and other turbine supervisory parameters. Generator winding temperature measurement. Measurement of electrical parameters like MVV. MVAR. Frequency. Excitation voltage etc. Condenser and feed tank level control, HP/LP Heaters level controls. Turbine seal steam pressure control. Turbine automatic run up and Synchronization. Grid coordinated operation of power plant.

**UNIT 4. Hydro and nuclear Power Plants:**

**A) Hydro Power Plant:** Types of water turbine. Regulation of speed and voltage. Surge tank level control.

**B) Nuclear Power Plant:** Concept of energy generated from atomic fission. Block diagram of an Atomic power station. Types of coolants. Control of chain reaction. Radio activity and safety measures.

**UNIT 5: Conservation of Electrical Energy:**

A) Measurement of Efficiency of power Plants, Criterion for selection of Instrumentation system / DCS system for power plant, Layout of control rooms.

B) Energy audit and conservation measures. Automation for conservation. Variable speed drives. Maximum demand and Power factor management. Use of soft starters.

C) Pollution control measures. Electrostatic precipitators. Boiler regulations.

**UNIT 6: Non-conventional Energy Sources:** Concept of power generation from non-conventional sources of energy like wind power, Solar Power and Tidal waves. Photovoltaic cells, Hydrogen cells. Power generation using incinerators and bagasse fired boilers.

**Experiments**

*Any 8 experiments based on above syllabus.*

**This should include:**

1. Visit to a power plant. Its study and report writing.
2. Study of power plant simulation technique.
3. Sequential start-up and shutdown of auxiliaries and related process drawings.

**Reference books**

1. Handbook of Instrumentation and Control By Kallen
2. Power plant Engineering By Fredrick Morse.
3. Process Control Instrumentation By F. G. Shinsky.
4. All the volumes on Power Plant Engg. Published by Central Electricity Generation Board of U. K.
5. O & M manuals of power plant prepared by Bharat Heavy Electricals Ltd.

**(Elective II)**

## 406272 (3): FIBER OPTIC INSTRUMENTATION

| Teaching Scheme (Hrs/weeks) | | Examination Scheme (marks) | | |
|---|---|---|---|---|
| **Lectures** | **Practical** | **Paper** | **Term work** | **Oral** |
| **4** | **2** | **100** | **25** | **50** |

**UNIT 1**

**Light and Waveguiding:** Nature of light, Waveguiding principles, dielectric waveguide total internal reflection, evanescent wave, acceptance angle, numerical aperture, skew rays, single mode fibers, types and classification of fibers, special fibers.

**UNIT 2**

**A. Transmission characteristics of optical fiber:** Attenuation, Material absorption losses, scattering losses, bending losses, intramodal and intermodal losses, overall fiber dispersion, and dispersion modified losses, polarization, nonlinear phenomena.

**B. Optical fiber measurements:** measurements of attenuation, dispersion, refractive index profile, fiber cutoff wavelength, numerical aperture, OTDR.

**UNIT 3**

1. **Optical sources for optical fiber:** Lasers, LEDs
2. **Optical detectors for optical fiber:** PN diode, pin diode, avalanche diode.
3. **Optical fiber connection:** Fiber alignment and joint loss, splices, connectors, couplers.

**UNIT 4: OPTICAL FIBER SENSING PRINCIPLES AND TECHNIQUES I**

A. Introduction to fiber optic sensing: Advantages and disadvantages of FOS, **Transduction technique based on intensity modulation:** evanescent field, coupling, encoding based position sensors.

**UNIT 5 OPTICAL FIBER SENSING PRINCIPLES AND TECHNIQUES.II**

A. Fiber grating technology and Fiber Bragg grating interrogation techniques. B. Distributed Optical Fiber Sensing.

**UNIT 6 OPTICAL AMPLIFICATION AND INTEGRATED**

**OPTICS:** Optical amplifiers, fiber amplifiers, integrated optics, integrated optical device beam splitters, directional couplers and switches, modulators, polarization transformation and frequency translators, optoelectronic integration.

**LIST OF EXPERIMENTS**

1. To study attenuation losses in optical fiber.
2. To study dispersion losses in optical fiber.
3. To study different splicing techniques.
4. To study OTDR.
5. To study characteristic curves of optical sources and detectors.
6. To measure numerical aperture of an optical fiber.
7. To study optical power meter.
8. Paper designing of an optical sensor.

**REFERENCES**

1. Optical Fiber Sensing Technology, Jose Miguel Lopez-Higuera, John Wiley & Sons, 2002.
2. Optical Fiber Sensors, John Dakin and Brian Culshaw, Artech house, 1997
3. Optical Fiber Communications, John M. Senior, Prentice Hall of India, 2nd edition.
4. Optical Fiber Communications, Gerd Keiser, McGraw Hill, 3rd edition, 2000
5. Electro-Optics Handbook, Ronald W. Waynant and Marwood N. Ediger, McGraw Hill, 2nd edition, 2000.
6. Fiber Optics Communications, Harold Kolimbris, Pearson Education, 2004.

## (Elective II)

## 406270 (4) PROCESS MODELING AND OPTIMIZATION

| Teaching Scheme(6Hrs/week) | | Examination Scheme (marks) | |
|---|---|---|---|
| **Lectures** | **Practical** | **Theory** | **Oral** |
| **4** | **2** | **100** | **50** |

**UNIT-1 MODELING AND SIMULATIONS**

**Introduction.** Types of models, modeling of process control systems in time domain and frequency domain. Fitting polynomials in the step test data. Language interpolation formula, least square fitting process models of some typical systems in differential equations form gravity flow tank. Tanks in series. Tanks in parallel dead time, first and second order models, higher order models. Modeling of first and second order electrical systems, mechanical systems, electromechanically systems and oscillatory systems.

**UNIT 2 MODELING OF MECHANICAL CHEMICAL SYSTEMS**

Reaction dynamics modeling the chemical reactors CSTR models plug flow reactor model modeling of flash drum distillation columns evaporators, driers, heat exchangers.

**UNIT 3 PROCESS IDENTIFICATION**

Identification of physical processes off line and on line identification. Step testing pulse testing sine wave testing ATV identification method, prediction error methods. Introduction to numerical algorithm for subspaces state space identification. Least square method. Relationships among time laplace and frequency domain.

**UNIT 4 ANALYSIS OF MULTIVARIABLE SYSTEMS.**

Open loop and close loop characteristics equations multivariable nyquist plot, loci plot, niederlinski index resiliency, interaction relative gain array (Bristol array) inverse nyquist array, robustnests doylt stein criterion, skogestad and morari method.

**UNIT 5 BASIC CONCEPTS OF OPTIMIZATION**

Continuity of functions Convex and Concave functions. Convex region, Extermum of the objective functions, quadratic approximation.

**UNIT 6 OPTIMIZATION OF UNCONSTRAINED FUNCTIONS**

Numerical methods for optimizing a function of one variable, scanning and bracketing procedure. Newton Quasi Newton and secant methods region elimination method; polynomial approximation methods. Multidimensional problem evaluation of unidimensional search methods unconstrained multivariable optimization, direct methods, indirect methods, secant methods.

***REFERENCE***

1. W.L Luyben, "Process Modeling Simulation and control for chemical engineers" by McGraw Hill 1973.
2. Thomas Edgar David Himmelblau – "Optimization of chemical process" second edition MeGraw Hill 2001.
3. W.F Stoecker "design of thermal systems" international Education Hill 1989.
4. J. Malley "practical Process instrumentation and control McGraw Hill.
5. Deo Narsingh, "System simulation with digital computer" prentice Hall India New Delhi.
6. Process Modeling And Optimization- Centre : Technical Coordination (C.T.C.) www.ctc.org.in

***LIST OF EXPERIMENTS***

* ***Perform any eight experiments from following***

1. Analysis of first /second order system by using step and ramp input.
2. Obtaining mathematical modeling of electrical/ mechanical system by first principle.
3. Obtaining mathematical modeling of liquid level system.
4. Study of distillation columns.
5. Study of Heat Exchanger.
6. Identification of second order process by prediction error method and

compare it with modeling by first principle.
7. Obtaining unknown parameters of second order process by least square technique.
8. Obtaining Relative gain array of any MIMO physical system.
9. Obtaining inverse Nyquist array of any Physical system.
10. Design of optimal control system by using quadratic approximation.
11. Analysis and comparisons of Quasi- Newton and secant methods.

## 406271: PROJECT

| Term work: 100 Marks | Oral: 50 Marks |
|---|---|

*The project selected should be such so as to ensure the satisfaction of the urgent need to establish a direct link between educational, national development and productivity, thus reduce the gap between the world of work and world of study. The term-work will consist of a report prepared by student on project allotted to them. The oral examination will be based on the above report and work.*

## 406272: INDUSTRIAL VISIT (Marks: 50)

The Institute should arrange 2 full days Industrial visit related to Instrumentation arid control industry. The students should submit visit report file with detailed information. Generally, students are expected to study following points in the industrial Visit.

- Process block diagram
- List of Process variables and control variables.
- List of Sensors and transducers and criterion for selection
- Control loops
- List of Control system components and process loop components
- Level of automation.

It is suggested that students should, identify the reasons for control strategy, implemented by the industry.

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